

Equations and Inequalities



Curriculum Ready

ACMNA: 234, 235, 236, 237, 240



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EQUATIONS & INEQUALITIES

Sometimes just writing variables or 'pronumerals' in mathematical expressions is not enough. Sometimes the value of the variable is needed. An equation is used to find the value of a variable.

Answer these questions, *before* working through the chapter.

I used to think:

What does it mean to say that an equation is 'linear'?

What does it mean to make a variable the subject of an equation?

What are the signs of inequality?

Answer these questions, *after* working through the chapter.

But now I think:

What does it mean to say that an equation is 'linear'?

What does it mean to make a variable the subject of an equation?

How are equations solved?



What do I know now that I didn't know before?



What is a Linear Equation?

An equation is a mathematical expression that has two sides separated by an equals sign (=) and at least one variable (or 'pronomeral'). If the highest power of the variable is 1 then the equation is called a 'linear equation'.

For example, these are all equations because '=' appears in each of them.

a $x + 4 = 12$

b $3x = 15$

c $2 + 3x = 8$

d $\frac{x}{3} = 5$

e $\frac{6x}{7} = 12$

These are also all **linear** because the power of x (the variable) is 1 (there is no x^2 or x^3 etc).

How are Equations Solved?

To solve ANY equation the goal is always to **get the variable by itself**. But whatever is done to find the variable by itself, must also be done to the other side. Each linear equation has only **one solution**!

Finding x by itself

If you look at the linear equation $x + 4 = 12$, it's probably easy to see $x = 8$. Your brain has actually simplified the equation to have x by itself without you even realising it.

This is how:

Subtracting 4 from the left side of the equation leaves x by itself. But the same must be done on the right side.

$$x + 4 = 12$$

$$\rightarrow x + 4 - 4 = 12 - 4$$

$$x = 8$$

For the equation in **b**, to find x by itself, both sides must be divided by 3

$$3x = 15$$

We must do the same to both sides

$$\rightarrow \frac{3x}{3} = \frac{15}{3}$$

$$x = 5$$

For the equation in **c**, it takes two steps to get x by itself

$$2 + 3x = 8$$

Subtract 2 from **both sides** to leave the term with x by itself

$$\rightarrow 2 + 3x - 2 = 8 - 2$$

$$3x = 6$$

Divide **both sides** by 3 to leave the x by itself

$$\rightarrow \frac{3x}{3} = \frac{6}{3}$$

$$x = 2$$

If the variable is in the numerator of a fraction, like in **d**, then both sides must be multiplied by the denominator.

For the equation in **d**, we need to multiply the left hand side by 3 to find x by itself

$$\frac{x}{3} \times 3 = 5 \times 3$$

$$x = 15$$

Sometimes the variable will be in a fraction. Multiply and divide both sides to get the variable by itself.

In the equation in **e** x is the numerator of a fraction

Multiply **both sides** by the denominator
(always remove the denominator first!)

$$\frac{6x}{7} \times 7 = 12 \times 7$$

$$6x = 84$$

Divide both sides by 6 to leave the x by itself

$$\frac{6x}{6} = \frac{84}{6}$$

$$x = 14$$

Sometimes the left side will have to be simplified by collecting like terms:

Solve for x in the following linear equation

$$2x + 5 + 3x = 30$$

Simplify by collecting like terms

$$\rightarrow 5x + 5 = 30$$

$$5x + 5 - 5 = 30 - 5$$

$$5x = 25$$

Divide both sides by 5 to leave x by itself

$$\frac{5x}{5} = \frac{25}{5}$$

$$x = 5$$

What happens if the Power of the Variable is 2?

If the highest power of the variable is 2 (ie. x^2 is in the equation), then the equation is not linear, but is called a **Quadratic** equation. To get the variable by itself on one side, the square root is used to change x^2 to x . However, the **square root of both sides** must be found. Each quadratic equation has **two solutions**!

Solve for x in the following linear equation

$$x^2 + 5 = 21$$

$$x^2 + 5 - 5 = 21 - 5 \leftarrow \text{Subtract 5 from both sides}$$

$$x^2 = 16$$

Find the square root of both sides

$$\rightarrow x^2 = (\pm 4)^2$$

$$x = \pm 4 \leftarrow \text{Because } 4^2 = 16 \text{ and } (-4)^2 = 16$$

$$\text{So } x = -4 \text{ or } x = 4$$

1. Choose whether the following equations are linear or quadratic by circling the appropriate word (don't solve):

a $x + 3 = 6$ is a linear / quadratic equation

b $3x^2 = 12$ is a linear / quadratic equation

c $3x = 12$ is a linear / quadratic equation

d $\frac{5x}{2} = 3$ is a linear / quadratic equation

.....
2. Solve these linear equations:

a $x + 4 = 7$

b $x + 3 = 10$

c $x - 2 = 13$

d $x - 6 = 10$

3. Find the value of the variable in each of these linear equations:

a $2x = 8$

b $5x = 35$

c $\frac{x}{4} = 5$

d $\frac{x}{3} = 8$

e $\frac{3x}{2} = 9$

f $\frac{5x}{3} = 5$

4. Solve for the variable in each linear equation:

a $4x + 3 = 23$

b $2a - 5 = 9$

c $5m + 6 = 31$

d $-4 + 7n = 24$

e $-2k = 10$

f $-m + 4 = 6$ (Hint: $-m = -1m$)

5. Solve for the variable in each linear equation:

a $\frac{y}{4} = 4$

b $\frac{5x}{4} = 15$

c $\frac{2p}{9} = 6$

d $\frac{4d}{11} = 8$

e $\frac{7x}{3} = -14$

f $\frac{10m}{4} = -25$

6. Solve these quadratic equations:

a $y^2 - 9 = 0$

b $4x^2 = 100$

What happens if the Variable is on BOTH sides of the Equation?

If the variables appear on both sides of the equation, one side needs to be changed to have no variables in it. REMEMBER! Whatever is done to one side must be done to the other side too.

Solve this linear equation:

$$8x - 4 = 3x + 6$$

3x can be subtracted from both sides. Now the right hand side will have no variables.

$$\rightarrow 8x - 4 - 3x = 3x + 6 - 3x$$

$$5x - 4 = 6$$

$$5x - 4 + 4 = 6 + 4 \leftarrow$$

Add 4 to both sides so that 5x is by itself

$$5x = 10$$

Divide both sides by 5 so that x is by itself

$$\rightarrow \frac{5x}{5} = \frac{10}{5}$$

$$x = 2$$

Solve this linear equation:

$$3p - 12 = 16 - 4p$$

4p can be added to both sides. Now the right hand side will have no variables.

$$\rightarrow 3p - 12 + 4p = 16 - 4p + 4p$$

$$7p - 12 = 16$$

$$7p - 12 + 12 = 16 + 12 \leftarrow$$

Add 12 to both sides so that 7p is by itself

$$7p = 28$$

Divide both sides by 7 so that p is by itself

$$\rightarrow \frac{7p}{7} = \frac{28}{7}$$

$$p = 4$$

If the equation has brackets, they must be expanded first. Then just solve the equation the same way as before.

Solve the following linear equation:

Expand both brackets

$$\rightarrow 3(x - 3) = 2(x - 2)$$

$$3x - 9 = 2x - 4$$

$$3x - 9 - 2x - 4 - 2x \leftarrow$$

2x can be subtracted from both sides so that the right side will have no variables

$$x - 9 = -4$$

Add 9 to both sides so that x is by itself

$$\rightarrow x - 9 + 9 = -4 + 9$$

$$x = 5$$

Equations with Fractions

Some shortcuts can be used when the variable is part of a fraction in the equation.

Type 1: One fraction in the equation – Multiply both sides of the equation by the denominator.

Solve the linear equation: (multiply both sides by the denominator)

$$\begin{aligned} \frac{x}{3} + 4 &= 6 \\ \text{The denominator of the fraction is 3} \\ \text{Multiply both sides by the denominator} \rightarrow 3 \times \left(\frac{x}{3} + 4 \right) &= 6 \times 3 \\ x + 12 &= 18 \\ x &= 18 - 12 \\ x &= 6 \end{aligned}$$

Here is another example with the variable in a fraction:

Solve the linear equation: (multiply both sides by the denominator)

$$\begin{aligned} \frac{(5x-3)}{2} &= 2x \\ \text{The denominator of the fraction is 2} \\ \text{Multiply both sides by the denominator} \rightarrow 2 \times \left(\frac{5x-3}{2} \right) &= 2 \times 2x \\ 5x - 3 &= 4x \\ 5x - 3 - 4x &= 4x - 4x \\ x - 3 &= 0 \\ x &= 3 \end{aligned}$$

Type 2: More than one fraction in the equation – Multiply both sides of the equation by the LCD of the fractions.

Solve the linear equation: (Find the LCD)

$$\begin{aligned} \frac{x}{3} + 5 &= \frac{x}{4} + 4 \\ \text{The LCD of the two fractions is 12} \\ \text{Multiply both sides by the LCD} \rightarrow 12 \times \left(\frac{x}{3} + 5 \right) &= 12 \times \left(\frac{x}{4} + 4 \right) \\ 4x + 60 &= 3x + 48 \\ 4x + 60 - 3x - 60 &= 3x + 48 - 3x - 60 \\ 4x - 3x &= 48 - 60 \\ x &= -12 \end{aligned}$$

Word Problems

Word problems can be transformed into equations which can be solved. Transforming the words into an equation is the most difficult part. The equations can be solved using one of the methods learned in this chapter.

When a number is halved the answer is 5, what is this number?

Let the number be x

According to the word problem $\rightarrow \frac{x}{2} = 5$

Multiply both sides by the denominator $\rightarrow 2 \times \frac{x}{2} = 2 \times 5$
 $x = 10$

In the above example $\frac{x}{2} = 5$ is the equation represented by the word problem.

To find an equation from a word problem, let a variable equal the missing value and use the information in the word problem to create an equation.

Daniel is 20 cm taller than Philip and the sum of their height is 320 cm. How tall is Philip?

- Let Philip's height be x cm
- So Daniel's height must be $(x + 20)$ cm

$$\text{Philip's Height} + \text{Daniel's Height} = 320 \text{ cm}$$

$$x + (x + 20) = 320$$

$$2x + 20 = 320$$

$$2x = 320 - 20$$

$$2x = 300$$

$$\frac{2x}{2} = \frac{300}{2}$$

$$x = 150$$

- So Phillip is 150 cm tall (and Daniel is 170 cm tall)

Lauren is 5 years older than her sister Maria. If the sum of their ages is 25, how old is Lauren?

- Let Lauren's age be x
- So Maria's age is $x - 5$

$$\begin{array}{cc} \text{(Maria)} & \text{(Lauren)} \\ \downarrow & \downarrow \\ \underbrace{(x - 5)} & + x = 25 \end{array}$$

$$2x - 5 = 25$$

$$2x = 25 + 5$$

$$2x = 30$$

$$x = 15$$

1. In these linear equations the variable appears on both sides. Solve for the missing value:

a $2u - 10 = 3u$

b $7x - 18 = 3x + 10$

c $3(x + 2) = -3$

d $4y + 18 = 12 - 2y$

e $10(n - 6) = 2(10 + n)$

f $6m - 4 = -5(m + 3)$

g $8(k - 4) - 5k + 3 = 4$

h $5(2y - 1) - 6(y - 2) + 3 = 6$

i $8t - (2t - 18) = -12$

j $2(a + 3) - 3(a + 4) = -10$

2. Find Ivan's mistake when he tried to solve this equation?

$$3(h + 2) = 2(h + 1) + 5$$

$$3h + 2 = 2h + 2 + 5$$

$$3h + 2 = 2h + 7$$

$$3h + 2 - 2h - 2 = 2h + 7 - 2h - 2$$

$$h = 5$$

3. Solve these linear equations which contain fractions:

a $\frac{x}{8} - 1 = 4$

b $\frac{n}{2} + 4 = \frac{5n}{12} + 5$

c $\frac{3b + 4}{5} = -4$

d $c + \frac{c}{2} = 12$

e $\frac{16r + 2}{5} = 10$

f $\frac{m}{2} + \frac{m}{3} = 5$

4. Solve these linear equations which contain fractions:

a $\frac{7q + 5}{3} = \frac{4q - 30}{6}$

b $\frac{5u + 5}{6} + \frac{5u + 10}{9} + 2 = 7$

c $\frac{2g}{3} + \frac{g}{10} = \frac{1}{2} + \frac{3g}{4}$

(Hint: Find LCD of ALL fractions)

d $2 = \frac{6}{x}$

(Hint: Multiply both sides by the denominator)

e $\frac{6}{8d - 2} = 1$

f $\frac{3k - 45}{k} = 2$

5. Three times a number is 45. What is the number?

6. Claire, Leanne and Lindsay are sisters. Claire is two years older than Leanne and Leanne is 4 years older than Lindsay. The sum of all their ages is 54. How old is each sister?

7. Charlie has been collecting stamps which he keeps in two separate books. The second book has 7 more than triple the stamps of the first book. If he has 35 stamps in total (from both books) then:

a How many stamps are in the first book?

b How many stamps are in the second book?

8. Victor has a bag filled with $2c$ and $5c$ coins. He has 2 more coins worth $2c$ than the coins worth $5c$. How many $2c$ coins does Victor have if all his coins sum to $88c$?

What are Linear Inequalities?

An inequality is a mathematical expression with two sides separated by one of these inequality signs:

- $>$ greater than
- $\geq$ greater than or equal to
- $<$ less than
- $\leq$ less than or equal to

For example $3 < 4$ and $8 > 2$.

If there is a variable in the inequality and its highest power is 1, then the inequality is a linear inequality (unless the variable is in a denominator). These expressions are all linear inequalities because an inequality appears in each of them and the highest power of the variable is 1.

- a $x + 2 < 5$ b $3x \geq 18$ c $-3x \geq 18$ d $\frac{x}{4} < 7$ e $2(x - 4) \leq 16$

How are Inequalities Solved?

Just like equations, the aim is to simplify the inequality to get the variable by itself on one side. Whatever is done to one side must be done to both sides.

Solve these inequalities

a $y + 3 \geq 10$

$$y + 3 - 3 \geq 10 - 3$$

$$y \geq 7$$

b $3m - 4 < 14$

$$3m - 4 + 4 < 14 + 4$$

$$3m < 18$$

$$\frac{3m}{3} < \frac{18}{3}$$

$$m < 6$$

Multiplying or Dividing by a Negative Number

Everyone knows that $2 < 5$ is true. If both sides are multiplied by -1 then $-2 < -5$. This is NOT true. If both sides of an inequality are multiplied by a negative number then the inequality sign must be reversed. So $-2 > -5$.

Solve these inequalities

a $-\frac{x}{2} > 5$

$$2 \times -\frac{x}{2} > 2 \times 5$$

$$-x > 10$$

$$x < -10$$

Inequality sign is reversed after multiplying both sides by -1

b $5 - 3x \leq 17 - 5$

$$5 - 3x - 5 \leq 17 - 5$$

$$-3x \leq 12$$

$$x \geq -4$$

Inequality sign is reversed after multiplying both sides by -3

Other than reversing the inequality sign when multiplying or dividing by a negative number, inequalities are solved in the same way as equations.

Variable is on both sides

a $5x + 25 \geq 2x + 4$

$$5x + 25 - 25 \geq 2x + 4 - 25$$

$$5x - 2x \geq 2x - 21 - 2x$$

$$\frac{3x}{3} \geq \frac{-21}{3}$$

$$x \geq -7$$

b $3m + 3 < 5m - 5$

$$3m + 3 - 3 < 5m - 5 - 3$$

$$3m - 5m < 5m - 8 - 5m$$

$$-2m < -8$$

$$m > 4$$

Inequality sign is reversed after dividing both sides by -2

Inequalities with brackets

a $3(q + 3) \leq 27$

$$3q + 9 \leq 27$$

$$3q + 9 - 9 \leq 27 - 9$$

$$\frac{3q}{3} \leq \frac{18}{3}$$

$$q \leq 6$$

b $2(y + 5) > 3(y + 5)$

$$2y + 10 > 3y + 15$$

$$2y + 10 - 10 > 3y + 15 - 10$$

$$2y - 3y > 3y + 5 - 3y$$

$$-y > 5$$

$$y < -5$$

Inequality sign is reversed after dividing both sides by -1

Inequalities with fractions (multiply by the LCD of ALL the fractions in the inequality)

a $\frac{6k - 8}{5} \geq 8$

$$5 \times \frac{6k - 8}{5} \geq 8 \times 5$$

$$6k - 8 + 8 \geq 40 + 8$$

$$\frac{6k}{6} \geq \frac{48}{6}$$

$$k \geq 8$$

b $\frac{t}{4} - \frac{2t}{3} \leq -10$

$$12 \times \left(\frac{t}{4} - \frac{2t}{3} \right) \leq -10 \times 12 \leftarrow \text{LCD of the fractions}$$

$$3t - 8t \leq -120$$

$$-5t \leq -120$$

$$t \geq 24$$

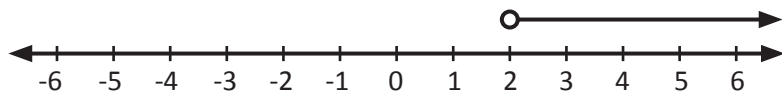
Inequality sign is reversed after dividing both sides by -4

Graphing Inequalities

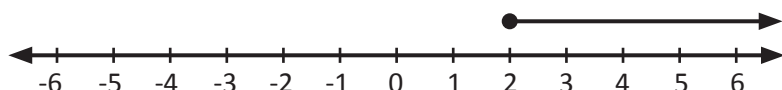
Solutions to inequalities can be represented on a number line.

For example, look at the inequality $x > 2$. This means x can be any number greater than, but not equal to 2.

On a number line $x > 2$ looks like this:



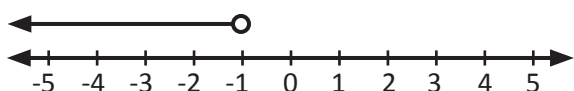
On a number line $x \geq 2$ looks like this:



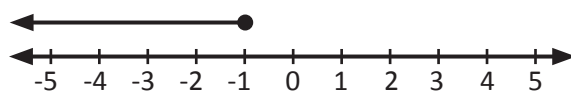
Can you spot the difference in the graphs above? In the first graph 2 is not included in the inequality ($>$), so the circle on the number line is hollow. In the second graph the inequality includes the number 2 ($\geq$) so the circle is solid.

Here are some more examples:

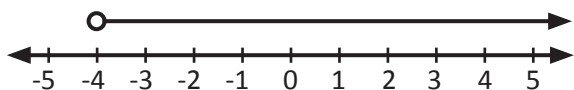
a $x < -1$



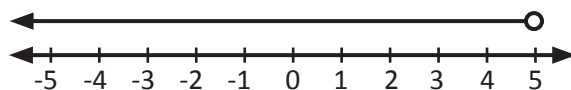
b $x \leq -1$



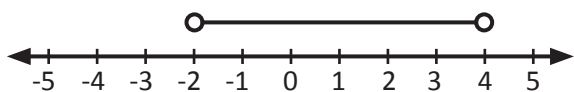
c $x > -4$



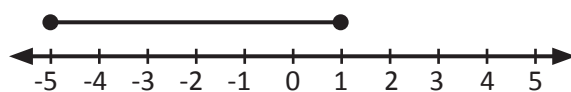
d $x < 5$



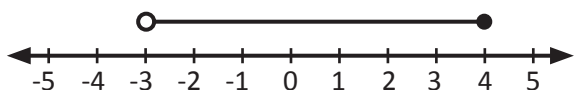
e $-2 < x < 4$



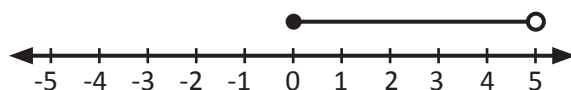
f $-5 \leq x \leq 1$



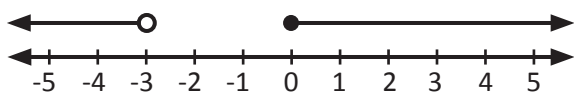
g $-3 < x \leq 4$



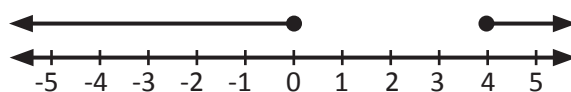
h $0 \leq x < 5$



i $x < -3$ or $x \geq 0$



j $x \leq 0$ or $x \geq 4$



1. Identify if the following are true or false:

a $6 > 3$

b $5 < 8$

c $3 > 8$

d $-2 < -5$

e $-4 > 4$

f $-8 < -4$

2. Solve these inequalities:

a $x + 3 < 4$

b $x - 4 \geq 5$

c $m + 7 \geq -4$

d $p - 10 \leq -8$

e $5q \leq 35$

f $4h + 3 > 51$

g $5x > 24 + x$

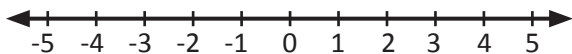
h $3(x - 1) \leq 3(1 - x)$

i $\frac{4h - 8}{2} < -2$

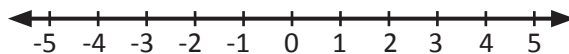
j $\frac{3x}{4} - \frac{2x}{5} \geq 14$

3. Graph these inequalities:

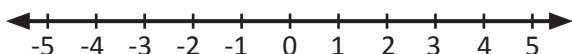
a $x > -3$



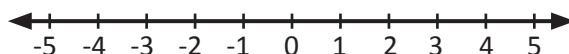
b $x \geq 2$



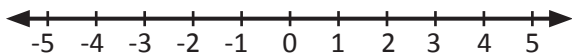
c $x \leq 0$



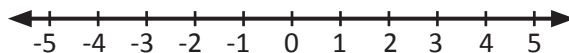
d $-1 < x < 3$



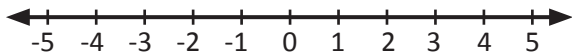
e $-5 \leq x < 5$



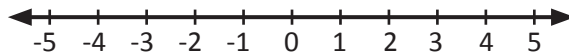
f $-2 < x \leq 4$



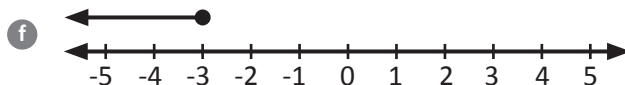
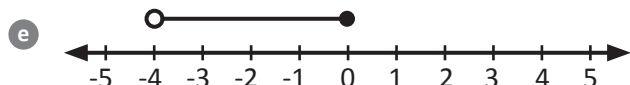
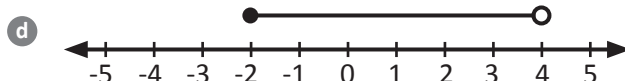
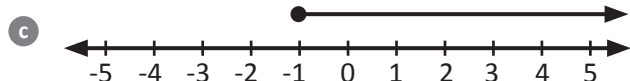
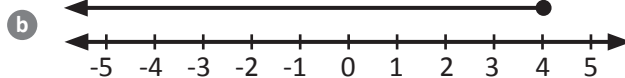
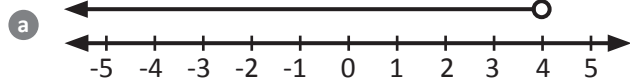
g $x < 0$ and $x \geq 3$



h $x > 4$ and $x \leq -1$



4. Write down the inequality represented by each of the following graphs:



5. Solve these more complicated linear inequalities, then graph their solution:

a $5x + 6 > 9(x + 2)$

b $7y - 4(y + 4) > 5$



c $\frac{4 - 3d}{8} \leq 2$

d $\frac{b}{8} - \frac{b}{3} \leq -5$



e $\frac{a}{-2} \geq \frac{4a + 3}{10}$

f $\frac{6c - 2}{48} < \frac{1}{2} - \frac{c}{12}$



Simultaneous Equations

What happens when there are two variables in equations?

The equation $x + y = 7$ has possible solutions:

$$x = 1 \text{ and } y = 6$$

$$x = -2 \text{ and } y = 9$$

$$x = 3 \text{ and } y = 4$$

$$x = 10 \text{ and } y = -3$$

and many other possibilities

The equation $2x + 3y = 18$ has possible solutions:

$$x = -3 \text{ and } y = 8$$

$$x = 3 \text{ and } y = 4$$

$$x = 6 \text{ and } y = 2$$

$$x = 10 \text{ and } y = -4$$

and many other possibilities

Common Solution

The equations have the **common solution** $x = 3$ and $y = 4$.

This solution solves both equations **simultaneously**.

Since two equations are solved at the same time they are called 'simultaneous equations.' There are three methods to solve simultaneous equations. All of them work with any question. It's up to you to choose the method you think is easiest.

Method 1: Substitution

In this method, one of the variables is made the subject of the formula and then substituted into the other equation. It has three easy steps.

Solve the simultaneous equations using substitution

$$x + y = 7 \quad \textcircled{1}$$

$$2x + 3y = 18 \quad \textcircled{2}$$

Step 1: Make one of the variables the subject in the equation, for example, using $\textcircled{1}$

$$y = 7 - x$$

Step 2: Substitute this expression for y into the $\textcircled{2}$ to make a new equation, and solve this new equation:

$$2x + 3(7 - x) = 18$$

$$2x + 21 - 3x = 18$$

$$-x = -3$$

$$x = 3$$

Step 3: Substitute this x -value into either of the equations to find the y -value:

$$\text{Substitute into } \textcircled{1} \quad 3 + y = 7$$

$$y = 7 - 3$$

$$y = 4$$

$$\text{Second equation } \textcircled{2} \quad 2(3) + 3y = 18$$

$$3y = 12$$

$$y = 4$$

So the solutions are $x = 3$ and $y = 4$.

As you can see in **Step 3**, it doesn't matter which original equation you substitute the value of the first variable into. The same answer is found for both. So choose the equation you think would be easier to use.

Method 2: Elimination

If the coefficients of one of the pronumerals are the same in both equations, then the equations can be subtracted from one another to eliminate one of the variables.

If the coefficients of both variables are different in each equation we can make them the same by multiplying one of the equations by an appropriate number.

Solve the simultaneous equations using elimination

$$x + y = 7 \quad (1)$$

$$2x + 3y = 18 \quad (2)$$

Step 1: Make sure one of the variables has the same coefficient in both equations

At this stage both x and y have different coefficients in the two equations.

If (1) is multiplied by 2 then the coefficient of x will be the same as in (2)

$$2 \times (1) \rightarrow 2x + 2y = 14 \quad (3)$$

same coefficient for x as (2)

Step 2: Subtract equations with the same coefficients

Subtract (3) from (2)

$$\begin{array}{r} 2x + 3y = 18 \quad (2) \\ -(2x + 2y = 14) \quad (3) \\ \hline y = 4 \end{array}$$

The x has been eliminated and we only need to solve for y

Step 3: Substitute the value of the solved variable into any equation to find the value of the variable which is still unknown

Substitute $y = 4$ into (1)

$$x + 4 = 7$$

$$x = 3$$

So the solutions are $x = 3$ and $y = 4$.

In **Step 3**, $y = 4$ could have also been substituted into (2) and the same value for x would have been found. So choose the equation you think would be easier to use.

Method 3: Graphical Method

If the straight line graphs are drawn for each linear equation, then the point where the lines intersect (x, y) will be the solution to x and y respectively.

Solve the simultaneous equations using the graphical method

$$x + y = 7 \quad \textcircled{1}$$

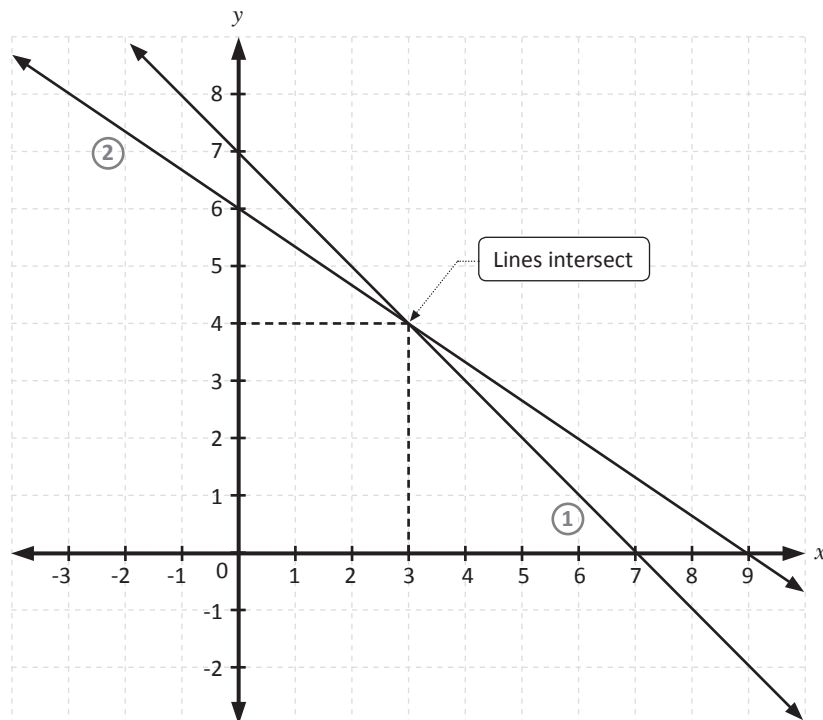
$$2x + 3y = 18 \quad \textcircled{2}$$

Step 1: Make y the subject of the equation for $\textcircled{1}$ and $\textcircled{2}$

$$y = -x + 7 \quad \textcircled{1}$$

$$y = -\frac{2}{3}x + 6 \quad \textcircled{2}$$

Step 2: Draw the graphs of these two equations on the same axes



Step 3: Read the point where the lines intersect

The lines intersect at the point $(x, y) = (3, 4)$

So $x = 3$ and $y = 4$

Notice that all three methods have the solution $x = 3$ and $y = 4$.

All three methods work all the time so just use the one you feel is the easiest. The next page demonstrates all 3 methods with another example.

Find the solution to these simultaneous equations:

$$2y + x = 3 \quad (1) \text{ and } 3y + 4x = 2 \quad (2)$$

Using **a** substitution, **b** elimination and **c** graphical method.

a Substitution

Step 1: Make one of the variables the subject of (1) $x = 3 - 2y$

Step 2: Substitute this expression into (2) and solve: $3y + 4(3 - 2y) = 2$

$$3y + 12 - 8y = 2$$

$$-5y = -10$$

$$y = 2$$

Step 3: Substitute this value into (1) or (2) to solve for the remaining variable: $2(2) + x = 3$
 $x = 3 - 4 = -1$

$$\text{So } x = -1 \text{ and } y = 2$$

b Elimination

Step 1: Make sure one of the variables has the same coefficient in both equations

$$4 \times (1) = 8y + 4x = 12 \quad (3)$$

(3) has the same coefficient for x as (2)

Step 2: Subtract equations with the same coefficients to eliminate a variable

$$\begin{array}{r} 8y + 4x = 12 \quad (3) \\ -(3y + 4x = 2) \quad (2) \\ \hline 5y = 10 \end{array}$$

$$\text{So } 5y = 10$$

$$\therefore y = 2$$

Step 3: Substitute the value of the solved variable into any equation to find the value of the variable which is still unknown

$$\begin{aligned} \text{Substitute } y = 2 \text{ into (1) to obtain: } & 2(2) + x = 3 \\ & \therefore x = 3 - 4 \\ & x = -1 \end{aligned}$$

$$\text{So } x = -1 \text{ and } y = 2$$

c Graphical

Step 1: Make y the subject of both equations

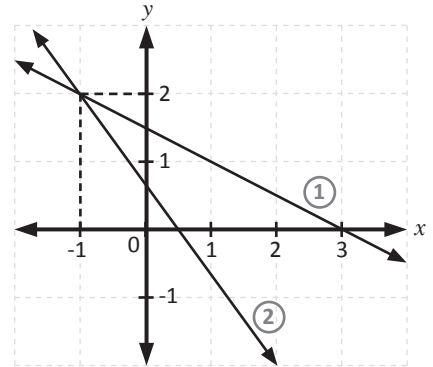
$$y = -\frac{1}{2}x + \frac{3}{2} \quad \textcircled{1}$$

$$y = -\frac{4}{3}x + \frac{2}{3} \quad \textcircled{2}$$

Step 2: Draw the graphs of these two equations on the same axes

Step 3: Read the point where the lines intersect

The lines intersect at $(-1, 2)$ so $x = -1$ and $y = 2$



As you can see, all three methods produce the same solution.

Simultaneous Equations Word Problems

As with single linear equations, word problems can be translated into simultaneous equations. Determine which TWO missing values are required and choose variables to represent these.

Write two equations using these variables and then use Substitution, Elimination or the Graphical Method to solve the equations.

The sum of two numbers is 12 and their difference is 6. Find the two numbers:

Let x and y represent the numbers.

So $x + y = 12 \quad \textcircled{1}$

and $x - y = 6 \quad \textcircled{2}$

These simultaneous equations can be solved using substitution, elimination or the graphical method.

Juan is twelve years older than his sister Jamila. In two years Jamila will be half Juan's age. Find Juan and Jamila's age:

Let x = Juan's age

Let y = Jamila's age

$x - y = 12 \quad \textcircled{1} \quad x + 2 = \frac{y}{2} \quad \textcircled{2}$

These are simultaneous equations which can be solved using substitution, elimination or the graphical method.

1. Write down 2 possible solutions for the variables in these equations:

a $x + y = 4$

b $2a + b = 6$

c $3x - 4y = 10$

2. Solve for the variables in these simultaneous equations using the substitution method:

a $2x + y = -1$

$$x - 2y = -4$$

b $2p + 3q = 10$

$$2q - 4p = 44$$

3. Use the graphical method to solve for these equations:

a $3x + 2y = 2$

$$2x - y = 6$$

b $3y - 4x = 24$

$$2y + 2x = 2$$

4. Solve for the variables in these simultaneous equations using the elimination method:

a $3x - y = -15$

$$y + 2x = 0$$

b $b - 4a = -12$

$$3a - 2b = -1$$

5. Solve these simultaneous equations using any method:

$$8c - 8d = 2$$

$$9c - 8d = 5$$

6. Solve these simultaneous equations using any method:

$$7m = 16 - 6n$$

$$2n = 3m + 16$$

7. Solve these simultaneous equations using any method:

$$6a - 2b - 10 = 0$$

$$2a + 3b - 29 = 0$$

8. Find equations and solve them for these word problems (using any method):

- a The sum of two numbers is 12. The sum of the first number, and double the second number is 16. What are the numbers?
- b Ari is three years older than Eric. In three years from now, Ari will be twice as old as Eric will be. How old are they now?
- c A restaurant sells two kinds of meals: pizza and pasta. A pizza costs \$14 and a pasta costs \$10. In a single day the restaurant sold 79 meals. If they earned \$994 on this day, how many of each meal was sold?

Basics:

- a Linear b Quadratic

c Linear d Linear
- a $x = 3$ b $x = 7$

c $x = 15$ d $x = 16$
- a $x = 4$ b $x = 7$ c $x = 20$

d $x = 24$ e $x = 6$ f $x = 3$
- a $x = 5$ b $a = 7$ c $m = 5$

d $n = 4$ e $k = -5$ f $m = -2$
- a $y = 16$ b $x = 12$ c $p = 27$

d $d = 22$ e $x = -6$ f $m = -10$
- a so $y = -3$ or $y = 3$

b so $x = 5$ or $x = -5$

Knowing More:

- a $u = -10$ b $x = 7$ c $x = -3$

d $y = -1$ e $n = 10$ f $m = -1$

g $k = 11$ h $y = -1$ i $t = -5$

j $a = 4$
- Ivan expanded the brackets incorrectly
Final answer should be: $h = 1$
- a $x = 40$ b $n = 12$ c $b = -8$

d $c = 8$ e $r = 3$ f $m = 6$

Knowing More:

- a $q = -4$ b $u = \frac{11}{5}$ c $g = 30$

d $x = 3$ e $d = 1$ f $k = 45$
- $n = 15$
- Leanne is $18\frac{2}{3}$, Claire is $20\frac{2}{3}$ and Lindsay is $14\frac{2}{3}$
- a 7 b 28
- Victor has 12 coins worth of $5c$ and 14 coins worth of $2c$.

Using Our Knowledge:

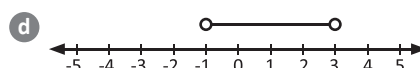
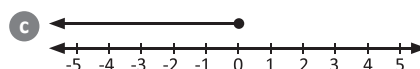
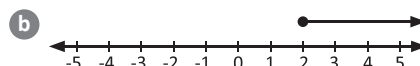
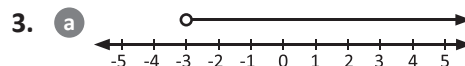
- a True b True c False

d False e False f True
- a $x < 1$ b $x \geq 9$ c $m \geq -11$

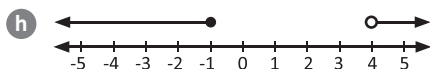
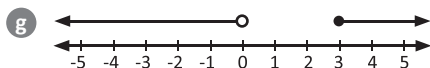
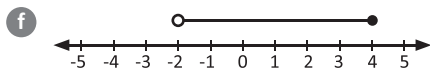
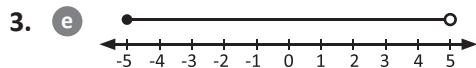
d $p \leq 2$ e $q \leq 7$ f $h > 12$

g $x > 6$ h $x \leq 1$ i $h < 1$

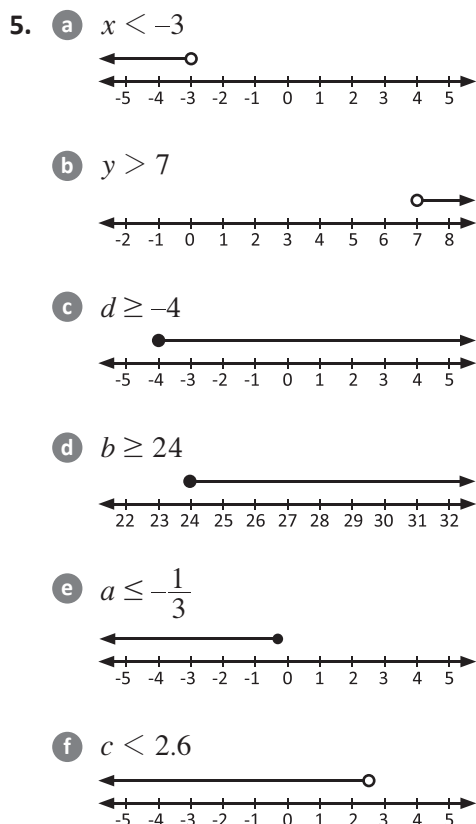
j $x \geq 40$



Using Our Knowledge:



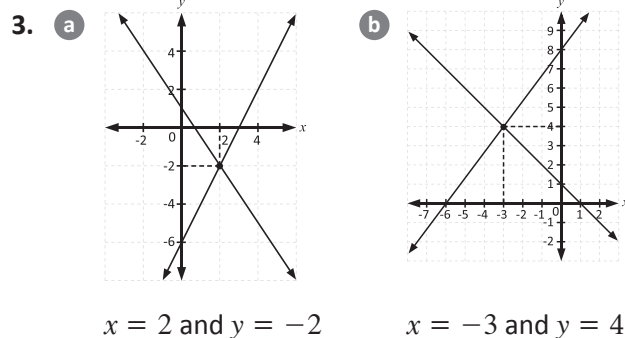
4. a $x < 4$ b $x \leq 4$
 c $x \geq -1$ d $-2 \leq x < 4$
 e $-4 < x \leq 0$ f $x \leq -3$



Thinking More:

1. a Anything where x and y add up to 4
 Eg $x = 3, y = 1$ or $x = 2, y = 2$ or
 $x = 4, y = 0$ or $x = 5, y = -1$ or
 $x = 3.5, y = 0.5$ etc
- b Anything where twice a , plus b , makes 6.
 Eg $a = 1, b = 4$, or $a = 2, b = 2$ or
 $a = 3, b = 0$ or $a = 4, b = -2$ or
 $a = 0.5, b = 5$ etc
- c Anything where 3 times the first number,
 minus three times the second number,
 makes 10.
 Eg $x = 6, y = 4$, or $x = -2, y = -4$ or
 $x = 30, y = 20$ or $x = 2, y = -1$ or
 $x = 1, y = -1.75$ etc

2. a $x = -1.2$ and $y = 1.4$
 b $p = -7$ and $q = 8$



4. a $x = -3$ and $y = 6$
 b $a = 5$ and $b = 8$

5. a $c = 3$ and $d = \frac{11}{4}$

6. a $n = 5$

7. a $a = 4, b = 7$

Thinking More:

8. **a** $a = 8$ and $b = 4$
- b** Eric is 0; Ari is 3.
- c** 51 pizzas and 28 pasta meals were sold



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